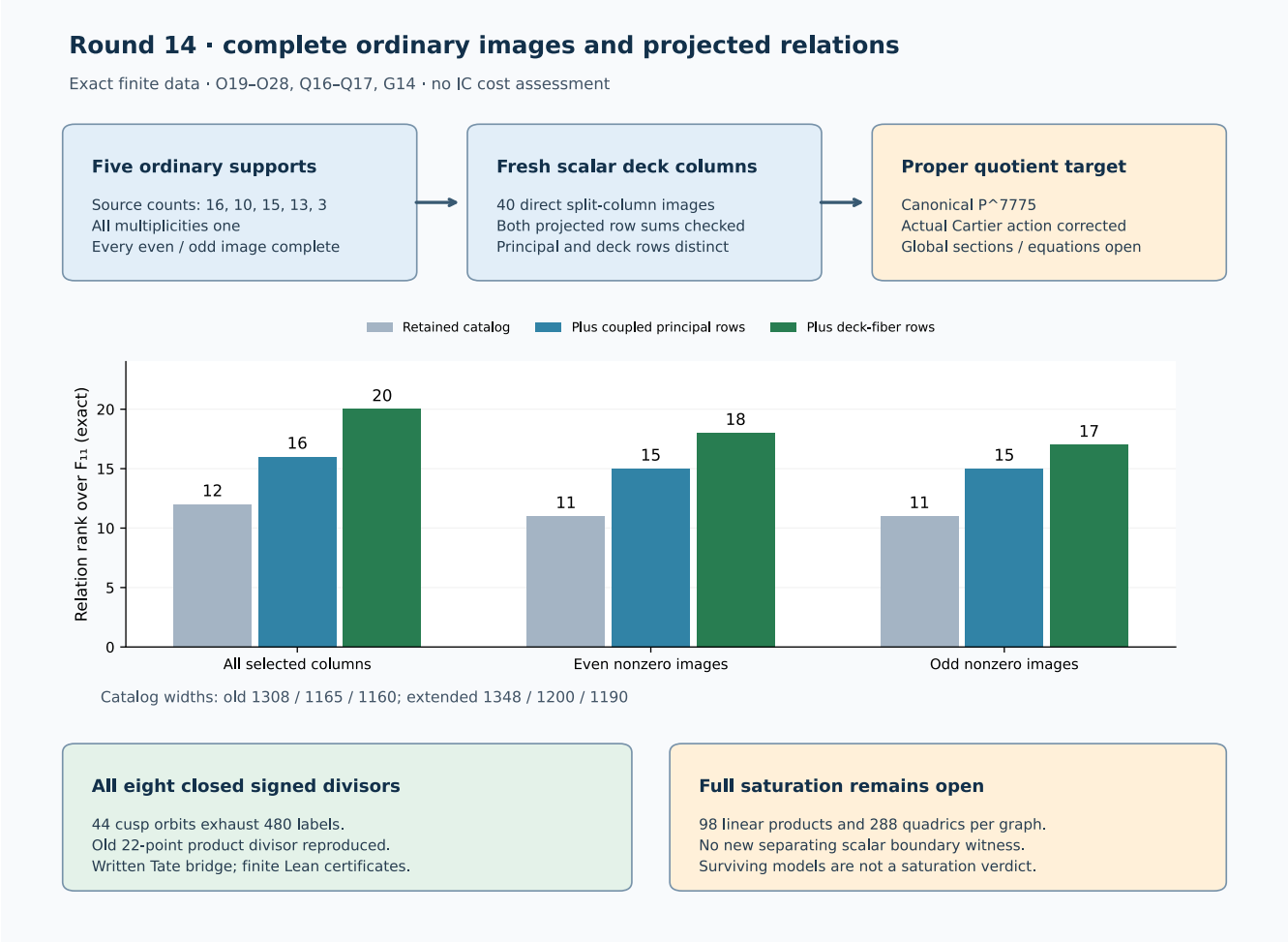
Round 14: complete ordinary supports and nonzero projected relations

20 October 2026, America/Los_Angeles. Parent3556fe3f6103bf212e18fcf7a37ac4acede8d6ed.

The five requested ordinary components now have57 simple closed points, with every even and odd image evaluated or proved zero under the retained written global bridges. Forty split columns include35 nonzero even images and30 nonzero odd images. The coupled principal rows raise the joined nonzero-image ranks from11/11 to15/15. Adding eight separate deck-fiber relations raises them to18/17. O28 completes all eight closed signed divisors under the written Tate/moduli bridge. Full quotient saturation and explicit proper polarized F2 equations remain open. No IC efficiency assessment is supplied.



Requirement status

Requested item	Status	Exact scope and remaining gap
Five ordinary supports and images	Verified complete under written bridges	57 simple closed points; all degrees, fibers and both projected images checked
Full quotient saturation	Partial	98 linear products and288 coefficient quadrics per graph; no new separating witness; full ideals/equality certificate absent
Proper polarized embeddings	Partial	Canonical6lambda target P^7775, actual pullbacks and Cartier character action; section bases, descent blocks and equations absent
Complete coupled divisors	Verified complete under written bridges	O28:44 closed cusps, retained SS points and all63 ordinary points; all eight signed divisors
Relevant Lean proofs	Partial	65 new named results and final axiom audit; global geometry and implementation semantics remain written

Five exact norms and all ordinary source points

The functions are x0+xinf+1, y0+yinf, y0+yinf+1, x0*yinf+xinf*y0 and x0*yinf+xinf*y0+1. The last two are cross-coordinate functions. They are not y-products. On E49: $y^2+xy=x^3+x^2+1$ over F2, cyclic121 Hecke norms recover A(x)+B(x)y at pole bounds504,756,756,780,780. All1024 coefficients match, using123904 input coefficients. The modular norm/good-reduction/pole/Riemann–Roch bridge is written; finite matching alone is not asserted to prove a global identity.

Function	Pushed ordinary degree	Pushed points	Source ordinary degree	Source points
x0+xinf+1	352	4	1760	16
y0+yinf	376	6	1880	10
y0+yinf+1	552	7	2760	15
x0*yinf+xinf*y0	274	5	1370	13
x0*yinf+xinf*y0+1	436	3	2180	3

O20 exactly factors every quadratic norm, reconstructs its products, checks Rabin conditions and resolves every y branch and common factor. Its cusp/SS/ordinary degree partitions match the independent round12 controls. All25 pushed ordinary points are simple. O21 enumerates all K-rational target and middle candidates. Actual cyclic kernels reject the noncyclic11 backtracks; modular-polynomial

compatibility alone is not used. O22 exports every scalar fiber, division factorization and commuting square. O23 reparses and checks each exported tower. O24 reconstructs all norms and pushed divisors again and exhausts the five source degree controls.

Seventeen source points each exhaust a degree5 deck fiber and have both images zero by the retained integral augmentation theorem. Eight other fibers split into40 points. The first degree280 base point, for example, gives five degree280 source points. Neither degree exhaustion nor their combined zero image determines the individual images.

Direct images and the relation-rank join

O25 evaluates the smallest split fiber: five degree7 points. Their even images are zero and their odd images are all nonzero. O26 completes the other seven split fibers in exact fields of degrees100 through560. Every basis has actual11-torsion, a Weil pairing aligned to the old F30 root and F32 encoding, all12 cyclic-line targets, and both distinguished targets checked. The five inverse-scaled basis changes per fiber preserve that pairing. The odd-sign F4 twist root is transported through the same old cyclotomic embedding. Every normalized Frobenius orbit, signed augmented trace, order11 condition and F32 descent passes. Across all eight fibers there are960 commuting-square checks and80 normalized-period checks.

O27 separately parses the final points into the prior u32 F32 arithmetic, checks curve membership, order11 and every projected row sum. Its integer and u64 eliminations agree. This is a separate native output checker with shared prior geometric formulas; no external CAS or independent-host audit is claimed. Exact replay reconstructs all exported arithmetic.

The40 columns are distinct from the prior1308: old closed degrees are 6,30,60,210,660, all divisible by6; new degrees7,11,28,33,50,85,280 are not. The eight new base-point keys and all scalar factors are also checked for distinctness. No equal-degree identification is guessed.

Joined rows	Evaluated rank / columns	Even nonzero-image rank / columns	Odd nonzero-image rank / columns
Retained round08 catalog	12 /1308	11 /1165	11 /1160
Plus coupled principal rows	16 /1348	15 /1200	15 /1190
Plus eight deck-fiber rows	20 /1348	18 /1200	17 /1190

The principal rows are complete after projecting away known zero images. The deck rows are separate projected divisor relations: each sums the five closed points above one base point, whose augmented deck norm vanishes. They are not labeled principal divisors. The native record exports the actual matrices and masks. Lean checks the new identity minors and their ranks, the finite F32 point/row calculations, disjoint-degree conditions and fresh-coordinate independence. The entire joined rank computation is native; no full Lean formalization of the elimination implementation is claimed.

Complete closed divisor assembly

Combining round10 cusp/pole valuations, round12 closed SS supports, round13 ordinary supports and this round's57 points completes every geometric signed divisor. All63 ordinary closed points across the eight functions now have both images. Cusp targets are14-torsion and SS targets are8-torsion; the retained11-primary projections kill them under the prior coprimality lemmas and written modular bridge. Every signed degree balances.

Function	Cusp zeros	SS zeros	Ordinary zeros	Poles
x0+xinf	420	1600	500	2520
x0+xinf+1	0	760	1760	2520
y0+yinf	360	1540	1880	3780
y0+yinf+1	0	1020	2760	3780
x0*xinf	2640	0	0	2640
x0*xinf+1	0	2060	580	2640
x0*yinf+xinf*y0	450	2080	1370	3900
x0*yinf+xinf*y0+1	0	1720	2180	3900

O28 enumerates normalized Frobenius on all480 actual cusp labels. Its44 closed orbits have degree histogram1:10,3:10,5:2,10:10,15:2,30:10 and reproduce the old22-point product divisor exactly. All target pairs, valuations and ramification indices are covariant. This completes all eight F2 closed signed divisors under the written Tate/moduli bridge in CUSPS.md. O27 retains its earlier open-cusp flag; O28 is the explicit refinement. The finite permutation and orbit coverage are in Lean.

Saturation and proper polarization

Q16 generates98 saturated linear product relations per actual circuit. Its147 coefficient queries per graph find no separating scalar model. Q17 expands all multiplication inputs and generates288 distinct boundary coefficient quadrics per graph. None of its288 queries separates the old restricted models. No query hits its100000-node limit. Nonzero boundary models still satisfy every repair; Lean checks them against the original restricted programs and all added relations. These surviving points do not prove incomplete saturation, and the restricted searches do not prove full saturation. The new Lean localization/contraction theorem states the full saturation criterion and the containment/h-regularity equality exit.

G14 binds the canonical Poincare construction to each actual graph quotient. Its very ample bundle represents6lambda_A, with7776 sections, projective target P^7775, Hilbert polynomial7776*t^5 and degree933120. Actual pullback matrices6*diag(H,[19]) have Euler characteristic2807136. The detailed written construction, primary source and missing executable steps are in GEOMETRY.md. No global section basis or projective equations were produced.

Review corrected the character action: the F19-linear dual is M^-transpose, whereas mu19-valued characters have Frobenius2*M^-transpose. The earlier linear-dual-only output is retained. Corrected character orbits have lengths 1,72,72,72,72,72 for each graph; all actual kernel images, additions and both pairings are checked. Lean proves the two actual Cartier matrix identities. Character permutations do not determine the semilinear section blocks. The existing six-chart Kummer atlas over F2^360 remains a prior result.

Failures, formal scope and validation

Both degree194 fibers hit the initial one-billion reduction cap; the frozen amendment raises O22 to five billion before successful retries. Wrong H catalog binding was rejected, then corrected to the actual graph catalog. Rust width/inference errors and failed Lean elaborations are retained. A full temporary filesystem interrupted linking and the first proof gate; task-specific scratch storage resolved that infrastructure failure. Final review also corrected the odd twist-root convention: choosing an arbitrary work-field root can negate an odd image relative to the old catalog. All unmatched-root exports and their replay are retained, then superseded by aligned exports, regenerated point proofs and final replay. The relation ranks and nonzero counts are unchanged by this correction.

The65 named Lean additions cover13 structural identities/criteria, eight actual boundary certificates, ten support arithmetic results and18 finite image/minor/independence results and16 cusp results. Universal polynomial claims have five source-bound IDC certificates. The complete final build and transitive axiom audit, including the exact total, are recorded in VALIDATION.md and ledger.json. No custom axiom, sorry or trusted native decision is admitted. Bit/field implementation semantics and the global modular, pairing and scheme bridges remain partially formalized, as PROOF_MAP.md specifies.

Raw evidence, failures, commands and hashes are retained in runs.tar.xz. Fresh extraction, exact replay and deterministic repack have separate validation receipts. Canonical frontiers are updated with these exact ranks and construction limits. No cost, rho comparison or IC efficiency verdict is supplied; that assessment remains with the user.

Evidence and formal boundaries

The raw runs.tar.xz archive and runs.sha256 retain native outputs and failures. REPLAY.sh rebuilds all mathematical outputs. ledger.json pins sources, raw inputs, identity certificates, requirement status and the final audit total. PROOF_MAP.md distinguishes literal kernel checks from written global geometry and native implementation semantics. GEOMETRY.md specifies the proper polarized target and the remaining section/equation construction.